

Gadgetize Update — Change Summary

local_mixing — 2026-08-20 → 2026-08-22

2026-08-22

What changed in the gadgetization step (the `gen_sandwich_gadget` single-carrier production path) across this update cycle. The companion design note with the full rationale is `docs/SWAP_REFRESH_REDESIGN`, the operational surface is `docs/GSS_MIX.md` (stage 1+2).

1 What was replaced

The aggregate-Gray fold era: time-invariant $[2, 2, 2, 3]$ masks whose cancellation in every fold's before/after XOR left the exact identity $\text{carrier}^+ \oplus \text{carrier}^- = \text{src}^+ \oplus \text{src}^-$ (100% of linear source gates), the Gray gather's linear operand-recovery witness, and a zero-slice guard at the input port only.

2 New mechanisms

1. **Symmetric junk-half port guards:** both ports carry independent draws of one slice-guard generator (identity exactly on the zero band slice, every nonzero slice perturbs the data, targets confined to the low forward-junk half).
2. **Per-gate mask swap-with-refresh:** at every fold the target and one random control each retire one base-degree monomial and gain a fresh draw (verbatim moves and polarity re-rolls are provably unsound); the target-side inject is strictly interior to the fragment stream.
3. **Gray declined; expanded fold + selective ladder (cap 3):** the gather materializes an operand's mask sum as an accumulator segment pair, so swap mode takes the expanded fold; the ladder re-spells wide fragments into `g57/CNOT`, borrowing scratch from live band variables.
4. **Target-stable commuting shuffle:** same-target XOR writes keep emission order, making the interior interleave exact.
5. **Relocation-coupled refresh:** every relocation refreshes one monomial of each moved value.
6. **Tail hygiene:** route-home before the strip (carrying the band placement map); every registry discharged (the telescope lemma — an undischarged registry breaks reverse evaluation); discharge helpers are live band variables via `loc`; fills source the low half.

7. **Split-channel epoch refills:** band-only linear pivot; carrier sources only inside degree-2 product terms.
8. **Peephole cleanup:** commutable identical pairs cancelled to a fixpoint (-2 to -3.5%); the deliberate mask redundancy is collision-guarded against any function-preserving local pass.

3 Contract changes

The composite preserves the sandwich on the *upper data half only* (payload contract; the closing guard perturbs the junk half), and is *reverse-honest*: the reversed gadget on $(a, 0, \text{zero band})$ yields $D^{-1}(a)$ on the upper half — one artifact evaluates C forward and D^{-1} backward. Both properties are verified at generation time. `PROD_SWAP=0` `PROD_CLOSE_SLICE=0` reproduces the prior Gray stream byte-for-byte.

4 Parameters and measured sizes ($n = 128$)

knob	default	effect
<code>PROD_SWAP/PROD_CLOSE_SLICE</code>	1/1	0/0 = legacy Gray
<code>PROD_LADDER_CAP</code>	3	1.62M g, 82% g57+CNOT (0: 883k/48% wide; 4: 3.3M)
<code>PROD_K</code>	3	$2 \rightarrow -18.5\%$ at $2\times$ piling-up ε
<code>PROD_CG_JITTER</code>	50	$0 \rightarrow -8.8\%$
both size levers		-27.3% : 1,174,130 full / 639,317 half

Baseline comparison: 1,616,025 full / 879,196 half versus the Gray build's 1,292,677 — $1.25\times$ for the full security upgrade.

5 Measured properties

`flip_match`: zero body matches in both orientations (residual: 0–2 exact tail-seam windows per seed at $\geq 99.4\%$ depth, plus occasional port-boundary fill coincidences). `segment_deduce`: zero interior-cut equations at degree 1 and windowed degree 2, both directions. Inputs $\leftrightarrow$ outputs affine rank full (257/257). Affine-predictor heatmap flat at the 0.5 ceiling. Forward and reverse verifies green at every size; gadget-family tests 140/140. Phase-A digestibility (100k-move server probe): splice rate 56.5% vs the 53.7% baseline, curated share 88.7% vs 87.3%, 6.28 bits/splice; the g57 share *rises* 46.6% $\rightarrow$ 58.5% through phase A; the compression channel succeeds on 2.1% of attempts.

6 Known residuals and cautions

The per-seed tail-seam window flicker (accepted; phase A rewrites that region first); `PROD_POST_FRAGMENT` re-opens broad linear leakage under swap mode (off-limits until audited); multi-carrier presets lack the output-port hygiene; mid-body refill balance is a measured property, not a theorem.